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Higher Market Thickness Reduces Matching Rate in Online Platforms: Evidence from a Quasiexperiment

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Abstract. Market thickness is a key parameter that can make or break a platform's business model. Thicker markets can offer more opportunities for participants to meet and higher chances that a potential match exists. However, they can also be vulnerable to potential search frictions. In this paper, using data from an online peer-to-peer holiday property rental platform, we aim to identify and measure the causal impact of market thickness on matching rates. In particular, we exploit an exogenous shock to market size caused by a one-time migration of listings from other platforms, which gives rise to a quasiexperimental design. We find that increased market thickness actually leads to lower matching rates. Keeping search technology and other factors constant, doubling market size leads to a 15.4% reduction in traveler confirmation rate and a 15.9% reduction in host occupancy rate. As a result, the platform lost 5.6% of potential matches because of the increased market size. We attribute the effect to increased search friction: travelers' search intensity increases by 18.3% when market size doubles. This effect is especially prominent when the matching needs to take place within a limited time. Our results offer insights for future empirical and theoretical research on matching markets. They also highlight that is important for platform owners to watch out for increased search frictions as markets grow and invest in search technologies to facilitate more efficient search.

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Keywords: matching markets • market thickness • matching supply and demand • quasiexperiment • difference-in-differences

1. Introduction

Do thicker markets have higher matching rates? Intuitively, they should. Size is critical to the survival and success of many matching markets. A larger number of participants on each side offers more opportunities for participants to meet and better chances that a match exists. In the recent rise of peer-to-peer marketplaces, platform owners have expended significant effort fueling growth. For instance, ride-sharing platforms, such as Uber, Lyft, and DiDi, spend billions of dollars a year on marketing campaigns and incentives for drivers and passengers (*Business Insider* 2015, Newcomer 2016), whereas other marketplaces, such as the holiday rental platform HomeAway, expand through aggressive acquisitions (Conner 2013).

A key difference between these peer-to-peer marketplaces and many traditional two-sided markets (such as the markets for credit cards, newspapers, and mobile phone service) is the existence of a high degree of heterogeneity—among market participants and among the goods or services that they offer or desire. In such

markets, search is essential to finding the right match. However, a market with heterogeneity is vulnerable to various forms of search frictions (Bimpikis et al. 2017, Fradkin 2017, Kanoria and Saban 2017, Arnosti et al. 2018, Cullen and Farronato 2018, Horton 2018). Heterogeneity increases as the market grows as does search friction. The presence of abundant options makes it more likely that a potential match exists, but it can be more difficult to find.

More often than not, a need for matching comes with a time constraint. Some matching markets, such as most labor markets, marriage markets, and markets for organ transplants, have implicit deadlines. Others have more explicit deadlines. For example, a traveler needs to find an accommodation before the intended day of travel. A homeowner needs to find a cleaning service before holding a party. A student needs to find a seat before the school year starts. With limited time to complete a match, the impact of search frictions becomes even more prominent. It is, therefore, unclear whether thicker markets will necessarily lead to higher

matching probabilities, especially in matching markets with heterogeneity and time constraints.

Existing theoretical literature typically assumes that matching markets exhibit increasing returns to scale or constant returns to scale (e.g., Diamond 1982, Pissarides 1990). Empirically, the majority of studies, almost all on labor markets, find evidence consistent with constant returns to scale. However, Petrongolo and Pissarides (2001, p. 409) refer to these findings as “tentative,” because most studies rely on aggregate time series or cross-sectional observations, and “no study conducts a careful test of increasing returns to scale by testing, for example, whether the matching rate improves when the total number of participants increases.”

This paper does exactly that. It studies the effect of market thickness on matching rates, exploiting an exogenous shock to market size. The fact that, in our data, some markets are impacted while others are not gives rise to a quasiexperimental design that allows us to make causal inferences. In particular, we use data from a leading peer-to-peer online holiday rental marketplace operating in Australia that has over 40,000 listings located in 3,200 areas and 77 regions. These data offer several advantages for studying the research question at hand. They cover a large number of geographically separated markets, which allows us to associate market characteristics, especially market thickness, with matching outcomes. They include information on inquiries and bookings for individual travelers, which allows us to study the property of the matching function at a micro level. Last and most important, the company’s business strategy offers us a unique opportunity to exploit an exogenous shock to market size: migrations of listings from two previously acquired platforms expanded some of the markets significantly over a very short period.

The context also offers several appealing features that ensure the validity of our identification strategy: (1) all new listings migrated without selections either by the platform manager or by listing owners; (2) the timing of the event was idiosyncratic in nature: it was neither publicly announced nor anticipated by market participants; and (3) careful examinations of both control and treatment groups show parallel trends before the event took place.

Unlike previous literature, we do not find evidence of increasing or constant returns to scale. Rather, we find evidence of diminishing returns to scale. In the marketplace under study, additional increases in the number of market participants lead to lower matching rates. Specifically, keeping search technology and other factors constant, when market size doubles, traveler confirmation rate goes down by 15.4% and host occupancy rate goes down by 15.9%. In total, the platform lost 5.6% of potential matches on each day due to the increased market size.

What drives the difference between our findings and those in prior research? Is it caused by differences in

context, data disaggregation, or identification strategy? To investigate this question, we conducted additional analyses at the market level exploiting cross-sectional variations alone, as in previous studies, but not the exogenous shock. Without the use of the exogenous shock, we find constant returns to scale as previous studies do. However, when using the exogenous shock as an instrument for market size, we again find decreasing returns to scale at the market level. In other words, the difference in our results is most likely driven by a better identification strategy rather than a unique setting or simply that we use using disaggregated data.

The decrease in matching rates can be attributed to increasing search frictions in markets with more participants. We find that, when market size doubles, the number of inquiries sent by travelers increases by 18.3% and the number of inquiries received by hosts increases by 19.6% keeping search technology and other factors constant. The travelers also end up paying higher transaction prices, and hosts end up receiving lower revenues. The impact of search friction becomes more severe when the deadline for matching is more imminent. When there is plenty of time left, an increase in search intensity does not affect matching probability and transaction prices. However, when the deadline is close in time, increasing search intensity leads to lower matching rates and higher transaction prices.

Our findings offer several implications for future research in matching. They emphasize the need for more careful empirical research to study the causal effect of market thickness on matching rates as well as carefully evaluate the properties of the matching functions used in modeling. They also highlight the need for theorizing the sources of market frictions, especially how deadlines affect matching rates.

Our findings also offer practical implications for rising peer-to-peer marketplaces. Although many marketplace operators attempt to expand their platforms through aggressive marketing campaigns or acquisitions, it may not be an effective strategy if not accompanied by better search functionality. As the market grows, search frictions also become more prominent. Those once-thick markets could become thin because of such frictions. It is, therefore, crucial for platform operators to invest in technologies, such as interactive decision aids, search engines and recommendation systems, while expanding in size. Sometimes less obvious choices, such as limiting the number of options, may achieve better outcomes. Moreover, it is especially important to facilitate efficient search processes when the need for matching is time constrained.

2. Related Literature

Our paper contributes to the growing theoretical and empirical literature that studies peer-to-peer marketplaces as well as the matching market literature in

general. In particular, our work is related to four streams of literature: (1) marketplace innovations, (2) effects of thickness in matching markets, (3) search frictions in online matching markets, and (4) matching supply and demand in perishable goods markets.

2.1. Marketplace Innovations

There is a burgeoning literature on innovative marketplaces fueled by increasing use of information technology and growing controls over the design, implementation, and operations of markets to better match supply and demand. Allon et al. (2012) show how different levels of involvement by the moderating firm can affect market outcomes. Several papers study how information affects participants' behaviors and market outcomes (Moreno and Terwiesch 2014, Parker et al. 2016), whereas others study how price is determined or its effect on market outcomes (Li et al. 2016, Cachon et al. 2017, Kabra et al. 2017).

2.2. Effects of Thickness in Matching Markets

In labor markets, there is a long-standing discussion on whether matching markets are governed by increasing, constant, or decreasing returns to scale. Early theoretical work typically assumed increasing or constant returns to scale (e.g., Diamond 1982, Pissarides 1990). On the empirical side, with only a few exceptions, most evidence converges on finding constant returns to scale (see Petrongolo and Pissarides 2001 and references therein). Note that this line of empirical research predominantly focuses on labor markets because of data availability. Most studies are conducted using aggregate data on vacancies and unemployment. More importantly, the empirical evidence is largely drawn from observational variations, either times series or cross-sectional, lacking clear identification strategies.

In contrast, we focus on a different context: online peer-to-peer marketplaces, which are often characterized by higher degrees of heterogeneity and need for more imminent matches. Moreover, we use individual-level search and matching data to control for many potential confounding factors that may also affect matching rates. Finally, we attempt to estimate the causal effect of market size, which to our knowledge, has not been done before in this literature.

Relevant to our work, Cullen and Farronato (2018) use individual-level data from an online peer-to-peer freelance marketplace to study the growth of online matching markets. They document evidence consistent with constant returns to scale but does not study this effect causally. Kabra et al. (2017) show evidence of increasing returns to scale using data from a ride-hailing marketplace, which exhibits less heterogeneity than the holiday rental marketplace that we study. Lastly, using data from an online business-to-business auction platform, Bimpikis et al. (2017) find that

consolidating the ending times of auctions increases the platform revenue by 6.5%, primarily because of bidder-participation friction.

2.3. Search Frictions in Online Matching Markets

This paper is closely related to theoretical and empirical work that studies search frictions in online matching markets. Compared with traditional marketplaces, online marketplaces, such as eBay, Uber, and Airbnb, substantially reduce costs associated with acquiring information. As a result, they present previously unavailable opportunities for buyers and sellers to meet, negotiate, and trade. Bakos (1997) shows that reduction in buyers' search costs can avoid substantial allocational inefficiencies in differentiated markets. Brynjolfsson and Smith (2000) show empirically that online markets have lower frictions on many dimensions compared with conventional stores. Parker et al. (2016) show that text message services lead to higher efficiency in India's agricultural markets. Kroft and Pope (2014) find that Craigslist, a platform that allows individual users to advertise jobs, house rentals, and items for sale, significantly lowered the apartment and house rental vacancy rate and the number of classified job advertisements in newspapers. Similarly, by allowing consumers to buy and sell used products, Craigslist has led to a 2%–6% annual reduction in municipal solid waste in markets that it entered (Dhanorkar 2018).

Even so, online marketplaces are not frictionless. Various forms of frictions still exist. First, there is still some amount of information asymmetry. Arnosti et al. (2018) find that lack of job applicant availability information can lead to inefficient equilibria due to wasteful search in equilibrium. Using data from Airbnb, Fradkin (2017) finds that 15% of the initial contacts sent by guests were rejected by hosts due to stale vacancy. Using data from an online freelancing platform, Horton (2017) documents evidence that employers often cannot discern the availability of prospective workers, which causes their invitations to be rejected and reduces the job fill rate. Dhanorkar et al. (2015) find that buyers' uncertainty about the products and sellers' commitments to transactions reduce the success rate of online exchange of material and waste.

Second, when information is abundant, individual users sometimes tend to engage in many searches and evaluations. This is not only costly, but it can also result in users avoiding making decisions altogether because of information overload. As Hagiu and Jullien (2011) show, platforms have incentives to induce consumers to search more than they would like. However, such behavior may have unintended consequences. Ghose et al. (2014) show empirically that providing more information during the decision-making process may lead to fewer consumer purchases because of information overload. Using data from a hotel search

platform, Koulayev (2014) estimates that it costs \$10–\$30 to search a page of results and attributes it to high cognitive costs of comparing various characteristics of newly found hotels.

Third, individuals may exert negative externalities on others and on the entire platform, which in turn, leads to inferior matching outcomes. Romanyuk (2017) describes a cream-skimming externality: when sellers have limited capacity, a seller may reject a low-value buyer and strategically wait for a high-value buyer. As a result, he remains available and attracts a fraction of subsequent buyers who otherwise would contact other sellers, resulting in lower total surplus. Using data from eBay, Nosko and Tadelis (2015) empirically show a form of reputational externality, where one disappointing transaction may cause a buyer to update his beliefs about the quality of all sellers on the platform.

Platforms, therefore, have a significant role to play in designing marketplaces with fewer frictions and higher efficiencies. First, platforms could offer interactive decision aids, such as filters, sorting mechanisms, and comparison tools, to create a customized search experience that better caters to individual preferences. Haubl and Trifts (2000) show with laboratory experiments that such decision aids reduce consumers' search effort and increase decision quality.

Second, platforms could design better and more targeted recommendation systems to help users discover a potential satisfactory match more quickly. Fleder and Hosanagar (2009) show that recommendation systems that explicitly promote diversity will enable better consumer-product matches. Ghose et al. (2012) develop a ranking system that ranks products based on their average surplus. De los Santos and Koulayev (2017) make use of information available at the time of the consumer request and develop a customized ranking system based on its assessment of the consumer's preferences and the expected consumer type. In contexts in which consumers themselves are uncertain about their preferences and learn as they search, Dzyabura and Hauser (2018) show that recommendations are more effective if they encourage consumers to search undervalued products and products with diverse attributes. Nosko and Tadelis (2015) show using a large-scale field experiment that prioritizing higher-quality sellers can reduce reputational externality and lead to higher buyer retention. Using eBay data, Dinerstein et al. (2018) find that a platform redesign promoting more relevant offers to consumers and prioritizing low price offers can help strengthen seller incentives to lower prices and hence, reduce price-related search frictions.

Third, platforms could develop feedback and reputation systems to reduce frictions caused by information asymmetry (Tadelis 2016). Using data from an online intermediary for software development services, Moreno

and Terwiesch (2014) show that participants are very responsive to the numerical reputation score and also, the unstructured reputational information. Using multiple field experiments conducted on Airbnb, Cui et al. (2019) show that reputation systems can effectively reduce matching inefficiencies caused by racial discrimination.

It is worth noting that sometimes restricting the amount of information disclosed or limiting the number of choices offered or actions available can actually lead to higher search efficiency and better matching outcomes. Ghose et al. (2018) show that a carefully curated digest of social media content during the early stages of consumer search can lead to a substantial increase in the conversion rate. Romanyuk (2017) finds that partial information disclosure can reduce the cream-skimming externality and lead to higher total surplus than full information disclosure. Kuksov and Villas-Boas (2010) and Arnosti et al. (2018) both show that restricting the number of choices may actually lead to better outcomes. Lastly, Kanoria and Saban (2017) show that, to mitigate search frictions, a platform should force the more selective side of the market to reach out first while disallowing the less selective side from doing so.

2.4. Matching Supply and Demand in Perishable Goods Markets

Perishability of the good or service and limited capacity play a vital role in moderating the success of a market. A large body of literature in revenue management and dynamic pricing is devoted to better matching supply with demand in perishable markets (see Talluri and van Ryzin 2005 for an overview). As the deadline approaches and inventory depletes, prices evolve dynamically, and the consumers that get served also change. Recent research by Li et al. (2014), Lederman et al. (2014), and Tereyagolu et al. (2017) studies empirically how such perishability affects consumer demand and how firms can better manage their depleting inventories in the airline, hotel, and theater industries, respectively. A key difference, however, between a peer-to-peer marketplace and a traditional market is that decisions are decentralized in the former. Overall, our study highlights the need for theory to better account for the effects of deadlines in matching markets.

3. Background and Data

This section describes the business model of the online peer-to-peer marketplace under study and in particular, how matching works in this market. We then discuss the details of the data and features instrumental to our study.

3.1. Background

We receive data directly from an online peer-to-peer platform for holiday rentals in Australia. The company granted us data access in 2015. Most of the regions that

it operates in are tourist destinations. For example, the Gold Coast, Melbourne, Byron Bay, Sydney, Brisbane, Great Ocean Road, and the South Coast are among its top destinations. A property listed on the network is usually an entire apartment or house rather than a room within it (by contrast, many Airbnb listings are of single rooms or partial units). In other words, a traveler will typically have the property all to himself during the contracted rental period rather than sharing it with the host. A typical traveler (or a traveling group) stays for four to five nights. Over one-third of travelers travel with children. Because of the nature of these trips, the property is not just a place to stay. It is usually considered by travelers to be an important component of the holiday itself. For such reasons, travelers tend to plan early and spend a significant amount of time communicating with prospective hosts before making decisions.

The matching process works as follows. Property owners or managers join the platform by paying an annual fee, which enables them to list their properties as rentals. Similar to other online peer-to-peer rental networks, such as Airbnb, listings typically include photos of the property, descriptions, and information on availability and prices. An interested traveler will send an inquiry to the owner of a listing after browsing its web page. The inquiry is a built-in function of the platform, which automatically sends a message to the owner/manager's account and alerts her. Such an inquiry usually includes information about the trip and the traveler and questions regarding the listing. Through these inquiries, travelers attempt to confirm availability, ensure that pricing and property features are as advertised, discuss preferred payment method, and acquire any detailed information unavailable on the web page. The platform does not offer instant booking as Airbnb does.

The owner/manager receiving the inquiry can then decide whether to reply and what to reply. She may answer the traveler's questions and meanwhile, ask for more information about the trip and the traveler himself. Because of the nature of the trips, owners are typically concerned about whether prospective guests will keep good care of the property during their stay and whether they will be able to pay on time. Therefore, in the communication, they attempt to acquire information regarding the purpose of the trip (e.g., bachelor party, family getaway, or business travel) and the background of the traveler or the traveling group (e.g., high school teenagers, college students, newly wedded couples, or retired seniors). Sometimes, the owner will also include her email address or phone number to continue conversation in a more personal and timely manner. Eventually, based on the correspondence between the traveler and the owner, the two parties will determine whether to confirm the transaction. After confirmed, we consider the traveler and the owner to be matched.¹ The two matched parties can

arrange payment offline or online (using the platform's payment system). In sum, the matching process facilitated by the platform is more formal and curated than Craigslist but not as instantaneous as Airbnb.

3.2. Data

We acquired data on listing features, traveler inquiries, traveler bookings, and owner bookings for the calendar year 2014. The listing data include all 150,000 listings that ever registered on the platform since its start in 2001, among which about 48,000 listings were still live by the end of 2014. The data set provides information regarding the initial listed date, the location of the property (region, area, and geocode), number of bedrooms, maximum number of adults, and about 40 features associated with the property. These features are selected by the host. They include property types (e.g., house, apartment, cottage), what trip purposes the property is best for (e.g., corporate conference, family holiday, romantic getaway), nearby attractions (e.g., beach, national parks), amenities (e.g., spa, wheelchair access), guest accommodation (e.g., child friendly, pet friendly), and others features, such as special offers.

The traveler inquiry and traveler booking data include 3.91 million inquiries made throughout 2014. For each inquiry, we observe the date when it is sent, the listing that it corresponds to, the destination of the trip, the check-in and checkout dates, the number of people (adults and children) on the trip, the daily price quoted, and whether it is confirmed. Among the 3.91 million inquiries, 350,000 (about 9%) were eventually confirmed.

Owners may also block out certain dates because of personal needs or for other reasons, and they do this dynamically. After a date is blocked out by the owner, it will appear as unavailable to travelers on the calendar. Following the convention of the company, we refer to these block outs as "owner bookings." The owner booking data include 88,000 owner bookings in 2014. For each owner booking, we observe when and for which travel dates the booking is made.

Data from this platform offer several advantages in studying the impact of market thickness on search intensity and market rates. First, the data cover a large number of markets relatively separate from each other geographically, which allows us to associate market characteristics and market size in particular with matching outcomes. In some other matching markets, such as eBay, such geographical separation is not possible. Second, in addition to the final matching outcome, we also obtain inquiry histories for all travelers in all markets, which allows us to measure travelers' search efforts and examine how market thickness affects search intensity and in turn, matching outcome. Fradkin (2017) is another study that uses guest inquiry data but only for one city. Third and perhaps, most importantly, the acquisition and later, the migration of listings provide us a unique

opportunity to exploit an exogenous shock to market thickness, which we will elaborate in the next section.

The data are not without shortcomings. Unfortunately, because of business constraints, the company was unable to provide the web search history of each traveler before he communicated with owners. We, therefore, only observe a traveler if he sends an inquiry to a prospective host. We will address this data-censoring issue in the empirical analysis in the next section.

4. Empirical Framework

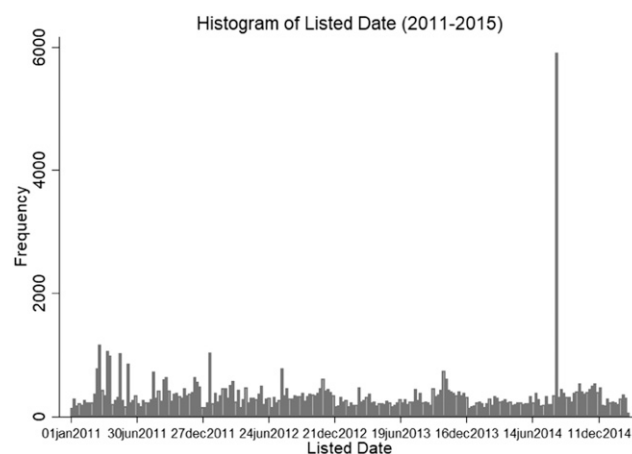
In this section, we introduce the exogenous shock that we exploit to identify the causal impact of market thickness on matching rates. We then describe the empirical specification for travelers and hosts. Lastly, we discuss the validity of this empirical framework.

4.1. Shock to Market Thickness

To identify the effect of market thickness on matching rates, we note that how thick a market will be is often determined endogenously. In two-sided markets, the number of participants on each side is affected by levels of demand and supply, demand-side and supply-side network externalities, and crossgroup externality (e.g., Rochet and Tirole 2003, Parker and Alstyne 2005, Armstrong 2006). Therefore, a direct cross-sectional comparison of thin and thick markets can be subject to many unobserved confounding factors that affect both market thickness and matching rates simultaneously. For example, higher demand levels or degrees of cross-group externality will lead to thicker markets (i.e., more participants on both sides). They also lead to higher matching probabilities, because participants have a stronger intention to find a match. Such factors, if present but not fully observed, can lead to a positive correlation and an upward bias in estimating the causal effect of market thickness on matching rates.

In this paper, we exploit an exogenous shock to market thickness to estimate its causal effect. In March 2011, the company acquired two competing peer-to-peer vacation rental platforms. After the acquisition, both acquired platforms operated under the original domain and brand names for more than three years. In August 2014, the company migrated listings from the two acquired platforms and consolidated them with existing listings on its original platform. For duplicate listings, the original listing on the platform was kept. Nonduplicate listings were added as new listings to the platform. The consolidation led to a sharp increase in the number of listings available on the platform as shown in Figure 1. Although about 30 new listings are added to the platform on an average day, a total of about 6,000 new listings were added during the four days when the consolidation happened: August 19–22, 2014. After the consolidation, traffic also started to migrate to the consolidated platform. Within a month,

Figure 1. Histogram of Listed Dates



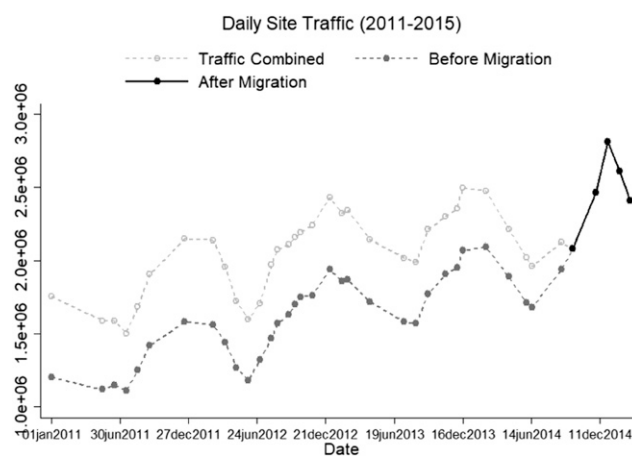
all traffic to the two acquired platforms was automatically redirected to the consolidated platform.

How did the consolidation affect the platform's market thickness? From the supply side, the estimated number of live listings on the platform just before the consolidation was around 42,000 (see Online Appendix A for details). After the consolidation, an additional 6,000 unique listings were added, representing a 14% increase. From the demand side, daily site traffic (i.e., number of unique users per day) rose about 13%–15% based on traffic data acquired from Alexa as shown in Figure 2. That is, the magnitude of the increase in demand is on par with the increase in supply. The shock, therefore, increased the thickness of the market while keeping the supply-demand ratio largely unchanged. It, therefore, provides us an opportunity to estimate the causal impact of market thickness on matching rates.

4.2. Difference-in-Differences Approach

Although the migration of listings provides us with an exogenous shock to market thickness, direct comparisons

Figure 2. Daily Site Traffic



of market performances between the before and after periods can result in biased estimates, because they can be confounded with time trends that are present even in the absence of the shock. For instance, Figure 2 shows that Australia's holiday rental market exhibits a clear seasonality. Demand is the highest during the summer months (December to February), which also include the Christmas and the New Year holiday seasons, and the lowest during the winter months (June to August). Because of this seasonality, the demand is expected to increase starting from September as more travelers book summer and holiday travels, even without the consolidation of listings. Other than seasonality, we also note that peer-to-peer holiday rental markets have in general enjoyed a steady growth in recent years, which is also shown in Figure 2, and this can be another confounding factor to the effect in which we are interested.

To isolate the confounding effects, our empirical strategy relies on identifying markets that did not experience the shock as the control group. These control markets experienced the same time trend, although they did not experience an increase in thickness of the market. We then compare the changes before and after for both the treatment and control groups.

In particular, we notice that not all markets experienced an expansion in size after the consolidation. For those markets that experienced an expansion in size, the magnitude was not uniform across markets; it varied substantially. We now describe how we define whether a market experienced a discontinuous increase in thickness because of the consolidation. We do not necessarily want to characterize the addition of any new listing as an indicator of an increase in thickness. Even without the consolidation, new hosts and new listings become available on a daily basis as shown in Figure 1, and the speed at which new listings join varies across markets. Therefore, we define a discontinuous change in market thickness as relative to the normal speed at which new listings become available in that market. Specifically, we compare the average number of new listings added per day during the four-day consolidation period with the number of new listings added daily during the nonconsolidation period. If the former is greater than the 90th percentile of the latter, then we consider that the consolidation has caused a discontinuous increase in market thickness.² We then measure the magnitude of the increase of market thickness as the ratio of the marginal increase of new listings during the consolidation period relative to the number of live listings just before the period (see details in Online Appendix A). The marginal increase of new listings is defined as the total number of new listings in the consolidation period minus the average number of listings in a typical four-day period.³

In sum, the possibility to construct control and treatment groups before and after the exogenous shock to market thickness leads to a quasiexperimental research design. For travelers, we estimate difference-in-differences models of the form

$$y_{irm}^{Traveler} = f(\alpha + \beta_1 Tr_m + \beta_2 A_{ir} + \beta_3 Tr_m \cdot A_{ir} + \gamma X_{ir} + \delta W_m + \epsilon_{jdm}), \quad (1)$$

where the subscript i denotes traveler, the subscript r denotes trip, and the subscript m denotes market (that is, an area in a region). Variable $y_{irm}^{Traveler}$ denotes the outcome variable (that is, whether a potential traveler is matched or not, number of inquiries sent, and price paid). The link function, f , concerns Logit, Poisson, and ordinary least squares (OLS), respectively. Binary variable Tr_m indicates whether a market m experiences a discontinuous increase in market thickness because of the migration of listings. One can also specify Tr_m as a categorical variable with multiple levels stratified by the magnitude of the increase or a continuous variable measured by the percentage of increase instead of a binary variable. A_{ir} denotes whether traveler i 's search for trip r is conducted after the shock (August 23, 2014).⁴ Matrix X_{ir} denotes traveler characteristics (e.g., country of origin) and trip characteristics (e.g., length of stay, weekend stay, number of adults and children, advance booking). Matrix W_m denotes market characteristics m (e.g., urban versus rural, business or family oriented). Error term ϵ_{irm} denotes the idiosyncratic shock to traveler i interested in trip r in destination m .

There are cases in which a traveler is flexible in his travel destination. As a result, we observe inquiries sent to listings in multiple destinations for the same trip. In such cases, we define treatment Tr_m and market characteristics W_m to be those of the primary market that the traveler is interested in, where the primary market is defined as the first inquired market for a trip. We choose the first market in which a traveler inquires as his primary market, because this market is likely to be given exogenously. Whether to search other markets can be influenced by the traveler's experience in the first market in which he inquires.⁵ To ensure robustness of the results, we vary the definition of the primary market to be the most frequently inquired market for a trip. Alternatively, instead of associating Tr_m and W_m with a single primary market of our definition, we define them as the average across all markets inquired. As we will show in the results, they are robust to these alternative specifications.

As we mentioned previously, our data are censored, because we do not observe when and where travelers intend to travel if they do not send any inquiry. Online Appendix B provides the details of how we correct for

data censoring when estimating the effect of market thickness.

For listings, we estimate difference-in-differences models of a similar form:

$$y_{jdm}^{Listing} = f(\alpha + \beta_1 Tr_m + \beta_2 A_{jd} + \beta_3 Tr_m \cdot A_{jd} + \gamma D_d + \xi_j + \epsilon_{jdm}), \quad (2)$$

where the subscript j denotes listing, the subscript d denotes day (or week), and the subscript m denotes market. Variable $y_{jdm}^{Listing}$ denotes the outcome variable (that is, whether a listing is occupied on specific stay date) and a listing's average daily revenue over a week (log scaled). We take the average to smooth the observed value, because many daily revenues are zero. The link function, f , concerns Logit and OLS, respectively. We define A_{jd} as equal to one if the calendar date d (or first day of the calendar week) is after August 23, 2014, and zero otherwise. For a listing date, the definition of "after" can be tricky, because even if a stay date is after August 23, 2014, it can be booked before August 23, 2014. Technically, these listing dates are not affected by the shock. Therefore, in our analysis, we exclude those listing-date combinations traveled after August 23, 2014, but booked before August 23, 2014, in both control and treatment markets. Note that excluding these listing-date combinations will lead to unbalanced samples for the treatment and nontreatment periods—whereas the nontreatment period includes all listing-date combinations, the treatment period will be overrepresented by listing-dates combinations that failed to attract early bookings. Therefore, to allow for fair comparisons, we also exclude listing-dates combinations during the nontreatment period that were booked before the start of the nontreatment period (i.e., June 18, 2014). Matrix D_d denotes stay date characteristics (i.e., day of week, holiday dummies). Error term ξ_j denotes listing fixed effect, and ϵ_{jdm} denotes the idiosyncratic shock to listing j on day (or in week) d in market m .

4.3. Validity of Difference-in-Differences Approach

In applying the difference-in-differences framework to the data, it is important to carefully consider the validity of the quasiexperiment created by the shock: migration of listings. Ideally, the treatment would be randomly assigned to markets. However, which market will experience an increase in thickness after the migration may not be completely random. In what follows, we investigate whether it can be endogenously determined or correlated with other factors that may affect matching rates.

First, all listings were migrated without filtering or selection by the company's management team. The decision to acquire these websites, which happened more than three years earlier, was driven by the corporate strategy to grow and expand in general rather

than to target a specific niche segment. Moreover, listing owners themselves did not select whether to migrate. Therefore, the typical selection bias that could threaten the validity of a quasiexperiment is not likely to be present in our setting. However, it is still possible that, although listing owners do not actively select whether to be migrated, they may passively choose to not engage with the new platform—because of unfamiliarity for instance. We consider this possibility carefully in our analyses and confirm that our results are consistent after accounting for it.

Second, although there is no strategic selection from either the management or user side, new listings are, after all, not randomly assigned to markets. Rather, markets with more presence from the two previously acquired platforms will experience greater increase in thickness, whereas others will experience less increase or none at all. If these are different types of markets, then the comparison between the control and treatment groups may be threatened. Luckily, the validity of the difference-in-differences approach hinges on the existence of the parallel trend. It does not require the control and treatment to be a priori the same in all aspects. It only requires that the two groups follow the same trend in the absence of the treatment, whereas the levels may differ because of differences in characteristics. This assumption can be investigated empirically using the data from the before period. In the subsequent section, we show that the parallel trend assumption is indeed satisfied, which supports the validity of our empirical method.

Other than these considerations, the shock that we exploit offers two other virtues. First, the change is sharp. Therefore, we expect a sharp and discontinuous change in the outcome variables, which can be visually examined to ensure that the effect is noticeable. Second, the timing of the consolidation was idiosyncratic, because it mostly depended on the capability of the information technology (IT) team and was neither part of the original acquisition plan nor preannounced publicly. Therefore, it is not likely that it was anticipated by listing owners or travelers.

5. Descriptive Results

In this section, we present the treatment and control groups; discuss how trip, traveler, and market characteristics compare between treatment and control groups; and examine the validity of the difference-in-differences framework.

5.1. Treatment and Control Groups

We define a market as an area within a region (recall that there are 3,200 areas and 77 regions in total). However, some of these areas are sparsely populated and rarely visited. We focus on areas with at least 35 listings just before the consolidation took place.⁶ We are thus left with 281 markets, which represent 83.7% of

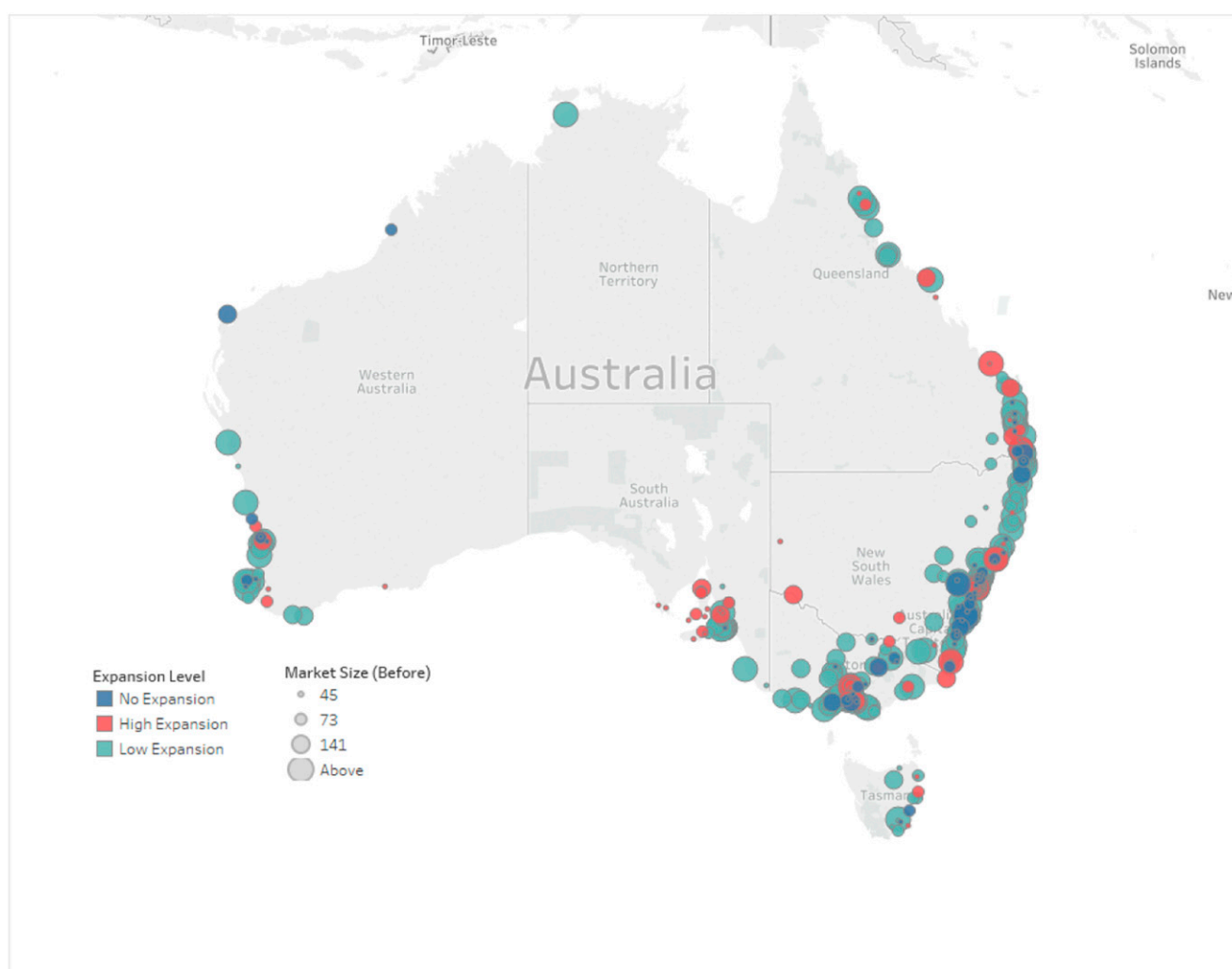
Table 1. Treatment and Control Groups Based on Levels of Increase in Market Thickness

	No. of markets	% Increase in market thickness	
		Mean	[5th percentile, 95th percentile]
No increase	108	0.0	[0.0, 0.0]
Increase	173	29.6	[2.2, 91.8]
Low increase	132	8.0	[1.8, 16.4]
High increase	41	34.6	[17.7, 62.7]
Total	281	13.6	[0.0, 62.3]

inquiries and 82.8% of bookings on the platform. Using the definition of treatment and control groups described in the previous section, we identify 173 markets that experienced discontinuous increase in market thickness postconsolidation, with an average percentage increase of 29.6% as shown in Table 1. For the purpose of illustrating comparability and parallel trends between treatment and control groups, we further divide the treatment sample into two subgroups: with low and high levels of increase

in thickness. However, we will account for the level of percentage increase in the subsequent empirical analysis. Using the 75th percentile of the percentage increase (17.1%) as the cutoff,⁷ we identify 41 markets with relatively high increase of thickness, with an average of 34.6% increase, and 132 markets with relatively low increase of thickness, with an average of 8.0% increase. As Figure 3 illustrates, the level of increase in market thickness is fairly evenly distributed across markets in

Figure 3. (Color online) Listing Expansions by Markets



different geographical locations and of different sizes. Online Appendix C presents examples of markets that experience different changes in thickness.

5.2. Comparability of Control and Treatment Groups

In this subsection, we examine whether the above construction of treatment and control groups leads to balanced samples of markets with similar trip, traveler, and market characteristics.

Our main analysis focuses on the period from 60 days before (June 18 to August 18, 2014) to 60 days after (August 23 to October 23, 2014) the migration of listings that took place in late August 2014.⁸ We use the rest of the data to inform us about the live status of each listing and dates that are blocked during the study window as well as for robustness analyses. During the four months of the study period, 940,000 inquiries were sent by travelers, with 83,000 of these eventually becoming confirmed matches. Note that a traveler may send out multiple inquiries for the same trip, sometimes with slight variations in destinations or travel dates. To accommodate such variations, we use similarities in destinations, travel dates, and search dates to group inquiries made by the same traveler for the same intended trip (see Online Appendix D for details). As a result, we associate 920,000 inquiries identified with traveler identifications with 456,000 trips. About 84%—that is, 771,000 inquiries and 372,000 trips—were in the 281 markets under study. Of these 372,000 trips, 56,000 (about 15%) were eventually confirmed.

To assess the comparability of treatment and control markets, we first compare trip characteristics and traveler characteristics between these two groups during the before period. We construct trip-level characteristics along the following dimensions: length of stay, weekend stay, number of weekends, number of adults, number of children, search days in advance, and holiday travel. To identify relevant holidays for the period that we study, we obtained 2014 and 2015 Australian national holidays from Australia.gov.au and complemented this information with the aggregated total inquiries for each travel date from our data. We, therefore, identify three holiday periods that are relevant for our analysis: Labor Day (October 3 to October 5, 2014), Christmas and New Year holiday season (December 20, 2014 to January 25, 2015), and Easter (April 3 to April 5, 2015). We create a dummy variable for each holiday that equals one if the travel dates contain any day during the holiday period and zero otherwise. The results are consistent if we define it as a continuous measure representing the percentage of travel days that fall into a holiday period. During the before period (i.e., two months before August 18, 2014), no inquiries were made for the Easter holiday in the coming year. Therefore, it is only relevant for the after period.⁹

Traveler-trip characteristics in Table 2 show that travelers and their trips are similar along all dimensions between treatment and control groups, and none of the differences in averages are statistically significant. Recall that there are cases in which a traveler sends inquiries to multiple markets. We assign the trip to the first inquired market by the traveler in these cases. Similarly, there are cases where travel dates may vary across inquiries within a trip. In such cases, we also define the travel dates of a trip as those of the first inquiry.

Next, we compare market characteristics between treatment and control markets. Although markets in the treated and control groups are similar along many dimensions, they are different along a few. We select the key characteristics among the 35 market characteristics based on *t* test results comparing means between treated and control groups and based on collinearity of these variables (see Online Appendix E for details). Market characteristics in Table 2 show the summary statistics for these key market characteristics by treatment and control groups. Markets in the treated group tend to be larger in size (that is, they have more listings). They also seem more urban (fewer rooms per property, more apartments or serviced apartments, farther away from adventures and national parks) and more business oriented (more executive listings, more listings with wheelchair access, more listings for short-term rentals). This suggests that it may be important to control for these variables when estimating the effect of market thickness on matching rate. Although these factors are likely already correlated with trip-level characteristics, such as length of day or weekend stay, they may also be correlated with other unobserved characteristics of the traveler or the trip such that it will affect how willing a traveler is to reach a deal.

5.3. Parallel Trend and Difference-in-Differences at Macro Level

In this subsection, we first evaluate whether the parallel trend assumption holds for the control and treatment groups before the migration of listings. We then examine whether it has led to a noticeable change in the number of inquiries sent by travelers, the number of listings that received inquiries, and matching rates in treatment markets relative to control markets.

Figure 4 displays these statistics for both treatment and control markets during the before and after periods. All data series are deseasonalized. Specifically, we use Hodrick–Prescott decomposition to remove day-of-week effects from the time series data. Figure 4 shows that treatment and control markets follow similar trends during the before period, a critical assumption to ensure validity of the difference-in-differences approach. Moreover, Figure 4 shows significant differences between treatment and control markets during the after period. The number of inquiries sent by travelers and the number of listings that received inquiries exhibit

Table 2. Traveler, Trip, and Market Characteristics by Treatment and Control Groups

Variables	Total	Control	Treatment		
			All	Low increase	High increase
Traveler-trip characteristics					
Length of stay	4.308 (4.321)	3.840 (3.786)	4.513 (4.521)	4.286 (4.095)	5.269 (5.651)
Weekend stay	0.833 (0.373)	0.820 (0.384)	0.839 (0.367)	0.834 (0.372)	0.856 (0.351)
No. of weekends	1.145 (0.513)	1.102 (0.489)	1.163 (0.522)	1.149 (0.515)	1.210 (0.542)
No. of adults	4.155 (2.892)	4.385 (3.025)	4.055 (2.826)	4.159 (2.919)	3.708 (2.460)
No. of children	0.912 (1.436)	0.971 (1.498)	0.886 (1.407)	0.903 (1.432)	0.828 (1.316)
Days in advance	63.53 (54.89)	60.22 (54.51)	64.99 (54.99)	65.03 (55.23)	64.85 (54.21)
Labor Day	0.0411 (0.198)	0.0402 (0.196)	0.0415 (0.199)	0.0426 (0.202)	0.0375 (0.190)
Christmas and New Year	0.160 (0.367)	0.151 (0.358)	0.165 (0.371)	0.167 (0.373)	0.157 (0.364)
Domestic	0.954 (0.209)	0.967 (0.179)	0.949 (0.221)	0.958 (0.200)	0.919 (0.273)
Market characteristics					
No. of listings per market	114.9 (85.87)	87.02 (46.77)	132.0 (99.20)	138.7 (99.23)	107.8 (97.58)
No. of rooms per listing	2.902 (0.544)	3.056 (0.441)	2.808 (0.581)	2.930 (0.520)	2.372 (0.592)
% of Apartment listings	0.146 (0.139)	0.0947 (0.0938)	0.178 (0.153)	0.142 (0.119)	0.306 (0.190)
% of Executive listings	0.232 (0.163)	0.189 (0.143)	0.259 (0.170)	0.246 (0.164)	0.304 (0.188)
% of Listings close to adventure	0.0195 (0.0240)	0.0236 (0.0313)	0.0169 (0.0178)	0.0177 (0.0180)	0.0140 (0.0173)
% of Listings close to national parks	0.263 (0.139)	0.287 (0.0938)	0.249 (0.153)	0.274 (0.119)	0.159 (0.190)

Notes. Summary statistics during the before period (June 18 to August 18, 2014). Standard deviations are shown in parentheses.

significant jumps right after the migration took place. We test for structural breaks with unknown break dates, and the results are displayed in Table 3. The test correctly identifies structural breaks on August 23 or 24, 2014 for the treatment markets right at the end of the migration period. The test did not identify any breaks with statistical significance for the control markets. After confirming that the event is real and impactful, we examine how the sharp increase in market thickness affected traveler- and listing-level matching rates. The lower panels in Figure 4 show that traveler confirmation rate and listing occupancy rate have both gone down for the treatment group during the after period relative to the control group. In Figure 4, the changes in markets with high levels of thickness increase are larger than for those markets with low levels of increase.

Although this analysis gives us the first evidence that increase in market thickness has led to lower matching rates, it does not properly adjust for trip, traveler, and

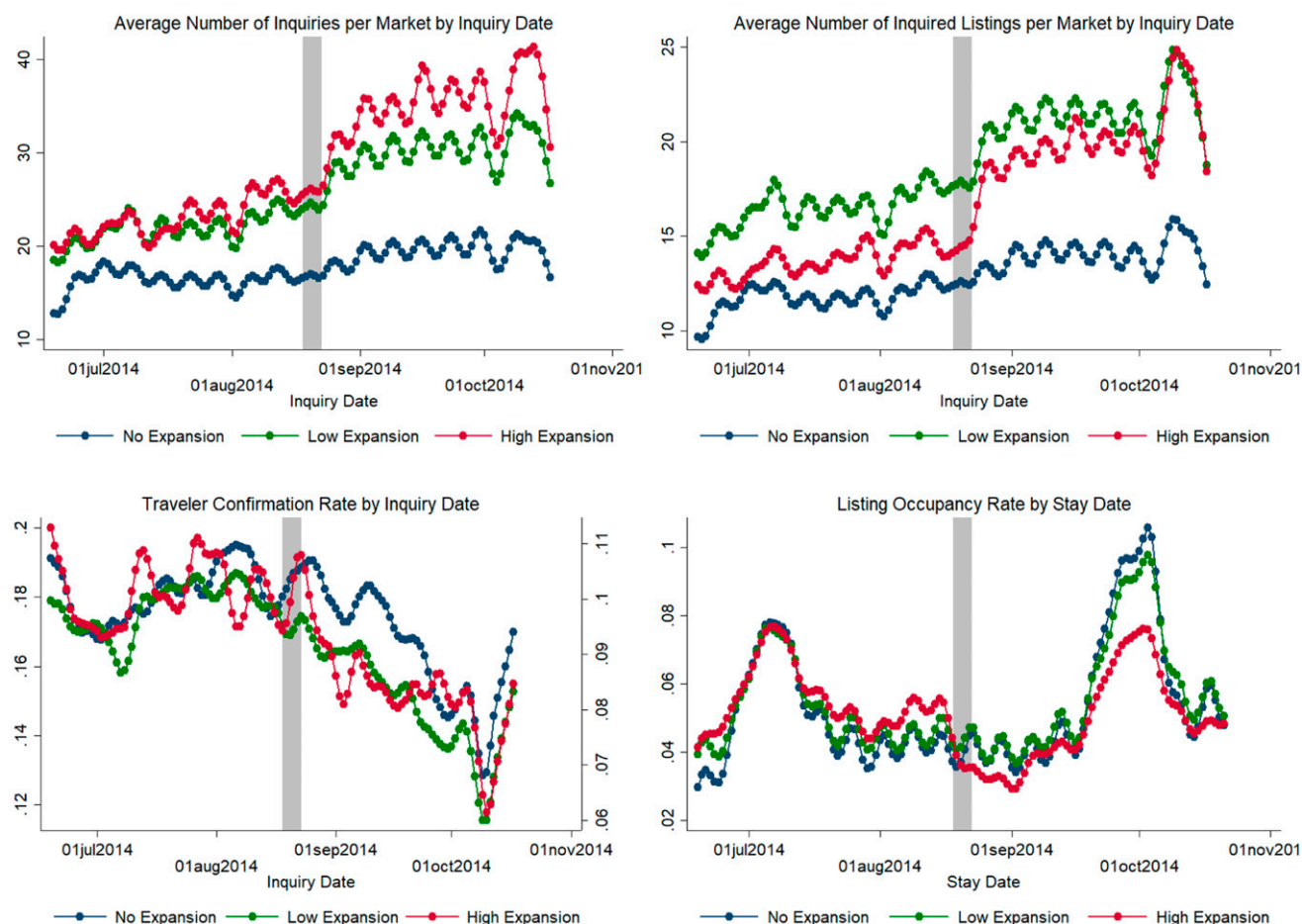
market heterogeneity or data censoring. In the next section, we will use the empirical specifications laid out previously to formally estimate the effect of market thickness on matching outcomes.

6. Estimation Results

In this section, we present the estimated effects of market thickness on matching probability and ensure that our results are robust to alternative specifications and explanations. We then compare our results with those in the literature using a similar market-level analysis to examine what drives the difference in the results.

6.1. Effect of Market Thickness on Matching Rate

Table 4 shows the estimated effect of increase in market thickness on matching probabilities. We first review the effects on traveler confirmation rate. Columns (1) and (2) correct for data censoring and report average semi-elasticities for easy interpretation. Column (1) shows

Figure 4. (Color online) Difference-in-Differences Comparisons

that a traveler's matching probability will decrease by 19.2% when market size doubles (i.e., increases by 100%). We calculate that, for an average traveler, his chance of being matched is 5.6% lower if his destination is among those markets that expanded after the consolidation. On an average calendar date, about 330 trips were confirmed in these markets. The 5.6% reduction in traveler confirmation rate, therefore, translates into a loss of 20 potential matches on each calendar date. Given an average daily rate of 290.95 Australia dollars (AUD) and an average length of stay of 4.59 days, the platform thus lost 26,709 AUD worth of transactions on an average calendar date.

Results in column (1) show that a traveler who visits the combined platform has a lower probability of being matched than a traveler who visits the original platform. One possible explanation for this lowered matching rate is that travelers on the merged and original platforms are different. For example, if travelers on the two merged platforms are more difficult to match to begin with, because they are either more selective or less desirable, the average traveler matching rate will be lower for the combined platform. Moreover, travelers from the two merged platforms could have a lower matching rate, because they are unfamiliar with the new system. To account for these possibilities, we conduct the same

Table 3. Test for a Structural Break with an Unknown Break Date

	No. of inquiries		No. of inquired listings	
	Identified break date	<i>p</i> -value	Identified break date	<i>p</i> -value
No increase in thickness	August 31, 2014	0.3833	August 23, 2014	0.4323
Low increase in thickness	August 24, 2014	0.0347	August 23, 2014	0.1844
High increase in thickness	August 24, 2014	0.0545	August 23, 2014	0.0341

Note. *p*-values are from supremum Wald tests.

Table 4. Effect of Market Thickness on Matching Rate

	Traveler confirmation rate			Listing occupancy rate		
	All TVLRs (1)	Existing TVLRs (2)	Existing TVLRs (3)	All LSTGs (4)	Existing LSTGs (5)	Existing LSTGs (6)
% Increase in thickness	−0.192*** (0.060)	−0.129* (0.069)	−0.154** (0.070)	−0.149* (0.080)	−0.145* (0.081)	−0.159** (0.094)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls			Yes			Yes
N	367,672	288,824	288,824	959,855	946,798	946,798
logL	−149,406.1	−120,595.9	−97,915.7	−210,166.9	−203,789.0	−207,168.5

Notes. Columns (1)–(3) report average semielasticities on treated markets during the after period. The estimates correct for data censoring. Standard errors are obtained through bootstrapping. Controls in these columns include main effects and trip, traveler, and market characteristics. Column (3) also includes average listing confirmation rate, average listing age (in days and log scaled), and percentage of listings joined during the four-day migration period as additional controls. Columns (4)–(6) report coefficient estimates from fixed effect logit models. Controls in these columns include main effects, listing fixed effects, and travel date characteristics. Column (6) also includes average number of prior trips that travelers have searched for on the platform as an additional control. TVLR, traveler; LSTG, listing.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

analyses on existing travelers from the main platform (that is, travelers who have sent at least one inquiry to some listings during the before period, which account for 74.3% of all traveler trips). The results are shown in column (2). Existing travelers also suffered from a lower matching probability after the consolidation. When market size doubles, the matching rate for existing travelers decreases by 12.9%. For an average existing traveler, his chance of being matched is lowered by 3.9% if his destination is among those markets that expanded after the merger. In Online Appendix F, we verify that our results are robust to variations in the definitions of primary market and market characteristics.

The results for all travelers and existing travelers both confirm that the increase in market thickness led to a lower matching rate for travelers. Next, we examine how listing occupancy rate changed as a result of the change in market thickness. It is possible that the lower matching rate for travelers is a result of intensified competition from the demand side. If so, the supply-side matching rate could actually go up. Columns (4) and (5) display the effect of market thickness on listing occupancy rate. Similar to travelers, listing owners also experienced a decrease in matching rate. As shown in column (4), the average occupancy rate decreases by 14.9% when market size doubles. A possible explanation for the lower occupancy rate is that migrated listings are more difficult to match. Another possible explanation is that the newly migrated owners are unfamiliar with the new system and may seem less responsive to travelers' inquiries. To test whether these possibilities could explain our results, we conduct the same analyses on existing listings from the original platform. As shown in column (5), when market size doubles, existing listings also experience a 14.5% reduction in occupancy rate. The results from the subsample analyses reassure that the lower matching rate is not simply driven by potentially lower matchability of the new participants from merged platforms. Because we do not have data on the two

merged platforms during the before period, these subsample analyses are critical to ensure fair before and after comparisons.

We further note that, in a two-sided market where each participant on one side can freely interact with any participant on the other, it is almost inevitable that some existing travelers would have interacted with a new listing owner migrated from the other two platforms or that some existing owners would have interacted with a new traveler as well. Moreover, even if an existing traveler has not directly interacted with a listing owner from the merged platforms but if the listing owners that he contacted have interacted with other travelers from those platforms, the traveler himself is indirectly connected to other participants from those merged platforms. If travelers and owners from the merged platforms are indeed more difficult to match, it may affect the matching rate of those participants who they directly interact with and maybe others through indirect connections as well (likely to less degree). It thus seems that the "cleanest" approach would have been finding a subset of participants (both sides) from the main platform that is completely isolated from the rest (no direct or indirect connections). This cleanest subset exists in our data, but it represents only 3% of all participants. This is not surprising, because if one had to restrict a significant portion of participants to interact with whoever they would interact with without the merger of the platforms, it defeats the purpose of acquiring and merging platforms. However, we do want to ensure that the observed result is not a result of some sort of contagious process but rather, that it is because of the treatment. We conducted additional analyses in which we control for characteristics of participants who an existing traveler or listing owner directly interacts with and see if the results would be significantly altered. The rationale is that, if the effects disappear or if their magnitudes are significantly smaller, then we might suspect that the lower matching rate can be a result of

Table 5. Effect of Market Thickness on Matching Rate: Market-Level Analysis

	Traveler confirmation rate			Listing occupancy rate		
	Before (1)	After (2)	After with IV (3)	Before (4)	After (5)	After with IV (6)
ln (no. of listings)	0.004 (0.008)	0.001 (0.007)	−0.031** (0.014)	0.000 (0.001)	−0.001 (0.002)	−0.006* (0.003)
Market characteristics	Yes	Yes	Yes	Yes	Yes	Yes
N	281	281	281	281	281	281
R ² (adjusted)	0.205	0.212	0.158	0.200	0.218	0.193

Notes. Columns (1), (2), (4), and (5) are estimated using OLS regressions. Columns (3) and (6) are estimated using 2SLS regressions. In columns (3) and (6), we use the number of new listings joined during the migration period (log scaled) as the instrument to the number of listings (log scaled) in the after period. Only existing traveler and existing listings are included in the analysis of traveler confirmation rate and listing occupancy rate, respectively.

* $p < 0.1$; ** $p < 0.05$.

lower matchability of new participants who went “viral.” The results are presented in columns (3) and (6) of Table 4. In column (3), in addition to previous controls, we also include average listing confirmation rate, average listing age (in days and log scaled), and percentage of listings joined during the four-day migration period as controls. In column (6), in addition to previous controls, we include average number of prior trips that travelers have searched for on the platform.¹⁰ As with columns (2) and (5), the analyses are conducted on the subsamples of existing travelers or existing listing owners only. We do not see any evidence of the effects going away, but they actually become more significant. This new evidence together with the previous evidence reassure us that the resulting lower matching rate is most likely due to the increased market thickness rather than differences of participants from different platforms if such differences exist.

In addition, we study heterogeneous treatment effects across markets of different sizes. We find that the effect of an increase in market thickness on matching rate is more salient in larger markets than in smaller markets. This is likely because larger markets offer more outside options, such as hotels and motels. Please refer to Online Appendix G for details.

6.2. Market-Level Analysis

Although the literature largely shows constant returns to scale in two-sided matching markets, we do not find evidence of increasing or constant returns to scale. Rather, we find evidence of diminishing returns to scale. We note that our setting and empirical method differ from those used in the literature in several significant ways. First, our context, peer-to-peer online marketplaces, is different from previous studies, which are primarily concerned with labor markets. It may be that different types of matching markets exhibit different returns to scale. Second, our study is conducted at the micro level (i.e., the matching outcome of each

trip or each listing day), whereas previous studies were conducted at the macro level (e.g., aggregate unemployment rate of a city or a country). Third, a more profound difference between our study and previous studies is that we exploit an exogenous shock to market size to make causal inferences, whereas previous studies rely on either time series or cross-sectional variations and lack clear identification strategies.

To understand the drivers of the differences in the results (context, aggregation, or identification strategy) and test the consistency of our results relative to previous ones under otherwise similar empirical specifications, we conduct analysis at the market level. We first examine whether a thicker market has a higher or lower matching rate, exploiting the cross-sectional variations at the market level as in previous studies. Next, to estimate the causal effect on matching rate at the market level, we again exploit the listing migration as an exogenous shock to market thickness. This time, we use it as an instrument to the number of listings in the after period and see if we will reach the same conclusions as previously obtained using the trip-level analysis or listing-day analysis.

Table 5 displays the results. Note that, when calculating market-level matching rates during the after period, we include only existing travelers or existing listing owners from the main platform. Based on the results shown in columns (1), (2), (4), and (5), one would reach a conclusion of constant returns to scale, consistent with the literature, albeit in a different context. Very interestingly, the results from the instrumental variables (IVs) regression in columns (3) and (6) once again point to decreasing returns to scale. That is, increases in market thickness lead to lower matching rates. This result not only shows the robustness of our findings, but it also indicates that the difference between our findings and those from the previous studies is not merely a result of differences in contexts or data aggregation but is likely a result of a causal relationship. The magnitudes of the

Table 6. Effect of Market Thickness on Search Intensity

	Inquiries sent per trip		Inquiries received per date		Quit success	
	All TVLRs (1)	Existing TVLRs (2)	All LSTGs (3)	Existing LSTGs (4)	Existing TVLRs (5)	Existing TVLRs (6)
% Increase in thickness	0.147*** (0.021)	0.183*** (0.021)	0.208*** (0.008)	0.196*** (0.008)	−0.117*** (0.029)	−0.308*** (0.090)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
N	367,672	288,824	3,283,400	3,156,276	683,706	683,706
logL	−637,834.0	−545,934.1	−2,904,629.8	−2,824,627.3	−2,874,054.7	−520,026.6

Notes. Columns (1) and (2) estimate effect of market thickness on the number of inquiries sent per trip by travelers using zero truncated Poisson regression. Columns (3) and (4) estimate the effect of market thickness on the number of inquiries received per listing date using Poisson regression with listing fixed effects. Columns (5) and (6) are estimated using Cox survival model. TVLR, traveler; LSTG, listing.

*** $p < 0.01$.

effects are comparable with those obtained from trip-level or listing-day analysis. Based on the estimates, when market size doubles, traveler confirmation rate decreases by $3.1 \times \ln(2) = 2.15$ percentage points, equivalent to a 14.0% reduction; listing occupancy rate decreases by $0.6 \times \ln(2) = 0.42$ percentage points, equivalent to an 14.2% reduction.

7. Search Friction and Market Congestion

Intuitively, one would probably expect that a higher market density leads to a higher matching rate. Our empirical results reject this hypothesis. We find evidence that increased market thickness may actually lead to worse matching outcomes, and we show this causally. In this section, we examine what drives this result.

First, we test whether the lower matching rate is a result of travelers exerting less search effort and quitting earlier after the consolidation. Table 6 displays the effect of market thickness on search intensity. Columns (1) and (2) show that travelers searched more (i.e., sent more inquiries per trip) in those markets that experienced an increase in thickness. Specifically, travelers' search intensity increases by 14.7% when market size doubles (i.e., increases by 100%). If we focus on those travelers who have had prior experiences on the main platform, their search intensity increases even more: by 18.3%. As a result, a listing owner receives 20%

more searches per stay date as shown in columns (3) and (4). To formally test whether a traveler is more (or less) likely to quit search after the merger, we conduct survival analyses on travelers' search process. Columns (5) and (6) show that travelers are less likely to quit search and also less likely to be matched after the merger. In sum, this evidence shows that the reduced matching rate is not a result of insufficient search effort. Rather, it could be a result of too much search, a possibility that we will elaborate shortly.

Second, although the matching rate is lower, perhaps the increased search effort leads to better matched outcomes. That is, perhaps travelers find lower-priced properties that match their needs. Or, perhaps listing owners receive higher revenues despite lower occupancy rates. Table 7 shows how the increase of market thickness affects prices paid by travelers and revenues received by listing owners. We do not find evidence of better matching outcomes. In fact, we find that travelers actually paid higher prices (9.8% higher) and listing owners suffered from lower revenues (7.9% lower).

How could travelers search more but match less and pay higher prices? Similarly, how could listing owners receive more inquiries from interested travelers but end up with lower occupancy rates and lower revenues? The lower matching rate is likely caused by search frictions in congested markets. When markets become

Table 7. Effect of Market Thickness on Price Paid by Travelers and Weekly Revenue of Listing Owners

	Paid price		Weekly revenue	
	All TVLRs (1)	Existing TVLRs (2)	All LSTGs (3)	Existing LSTGs (4)
% Increase in thickness	0.071* (0.039)	0.098** (0.044)	−0.094*** (0.030)	−0.079** (0.032)
Controls	Yes	Yes	Yes	Yes
N	50,529	41,192	577,248	542,529
R ² (adjusted)	0.318	0.317	0.00339	0.00348

Notes. Both price and revenue are log scaled. Columns (1) and (2) are estimated using OLS regressions. Columns (3) and (4) are estimated using fixed effect regressions with listing fixed effects. TVLR, traveler; LSTG, listing.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 8. Effects of Market Thickness by Closeness to Matching Deadlines (Existing Travelers Only)

Days in advance	No. of inquiries sent		Confirmation rate		Price paid	
	≤30 (1)	> 30 (2)	≤30 (3)	> 30 (4)	≤30 (5)	> 30 (6)
% Increase in thickness	0.286*** (0.036)	0.128*** (0.025)	−0.400*** (0.078)	0.076 (0.115)	0.263*** (0.071)	0.011 (0.056)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
N	111,163	177,661	111,131	177,661	16,353	24,839
logL	−192,929.0	−352,361.1	−46,802.2	−73,573.9		
R ² (adjusted)					0.231	0.332

Notes. Results are for existing travelers. Price is log scaled. Columns (3) and (4) show semielasticities corrected for censored data. Standard errors in columns (3) and (4) are obtained through bootstrapping.

*** $p < 0.01$.

denser, both parties face a greater number of potential options. For travelers, the number of options available is evident after they enter the trip parameters and land on the search results page. For listing owners, although demand is not directly observable to them, they will start to see an increase in their popularity after receiving multiple inquiries from travelers. Many studies have shown that individuals have difficulty making decisions when facing an abundance of choices (Jacoby et al. 1974, Iyengar and Lepper 2000, Bertrand et al. 2010). To understand whether this is likely the case in our context, it is helpful to take a closer look at the matching process. As we discussed in Section 3.1, the matching process is not instantaneous. Because of information asymmetry, travelers and listing owners need to spend time and effort to find out whether there is a potential match between what they need and what the other party can offer. During his communication with a listing owner, a traveler is trying to figure out whether the property offers features desirable for the intended trip; a listing owner is trying to ensure that the traveler is capable of making the payment on time and will properly maintain the property during the stay. Because of uncertainties, a traveler (a listing owner) could be communicating with multiple listing owners (travelers) simultaneously. Although both parties obtain information from each other and make comparisons, valuable time could have passed. When travelers are ready to decide, their favorite choice may no longer be available (due to information asymmetry), or the other party may not be ready to commit yet (due to information overload); therefore, they are forced to seek either pricier options or quit altogether and seek outside options. As a result, travelers either end up paying more for their trips or simply end up unmatched. An alternative explanation for higher transaction prices is that listing owners raised their prices after the consolidation. We find it is unlikely to be the case given that the quoted price by listing owners (conditional on receiving an inquiry) remains the same after consolidation. Because we do not observe prices

when there is no inquiry received for a stay date, we cannot fully examine this possibility, which also leaves open opportunities for future research. Yet, a third explanation for higher transactions prices is that higher search frictions prevented customers from fully exploring the market and finding lower prices. However, this would seem inconsistent with the finding that travelers sent more inquiries after the shock. Although we believe that this friction is likely what causes the matching rate to decrease when the market size increases, without access to data on the timing and content of the communication between travelers and listing owners, we have to leave it for future research to further examine the form and extent of this friction.

We expect this congestion effect to be more severe when there is less time to match. This is because if the need for match never expires, travelers could search indefinitely, and they eventually can find a match, because unmatched listing owners are still available. However, when a matching deadline is close, the time pressure will force travelers to give up more frequently and seek outside options. This is what we observe in Table 8. When the intended travel dates are less than one month ahead, doubling market size leads to a 28.6% increase in the number of inquiries sent, 40.0% decrease in confirmation rate, and 26.3% increase in transaction prices compared with only 12.8% increase in the number of inquiries and no change in confirmation rate and transaction prices when the intended travel date is more than one month away. This result confirms that, although increase in market thickness leads to higher search intensity for trips that are nearer or farther in the future, it only leads to worse matching outcomes when the matching deadline is close.

Finally, we would like to point out some caveats in interpreting these results. We should note that the results hold true when the matching process and the search technology remain unchanged as a market becomes thicker. If, for example, the platform makes the matching process more instantaneous (such as the instant booking function offered by Airbnb) or limits the

number of options presented to travelers through more refined search, it may be able to reduce search friction and market congestion, which presents an interesting area for future empirical research. Next, the observed effects are best interpreted as applicable for markets that reach a certain scale. When markets are sufficiently small, growth in market sizes may generate different effects. Our analysis focuses on markets that have at least 35 listings. However, for those markets in their early growth stages that have only a handful of listings, it is possible that search frictions do not play a dominant role, and increase in market thickness could lead to better matching outcomes. Last, our results show that higher market thickness can lead to lower matching rates for participants on both sides of the market, but it does not necessarily indicate lower overall matching efficiency. Lower matching rates mean that participants are less likely to find successful matches, but those who do may have found better matches. The higher average transaction price may be an indication that listings are matched to travelers who value them more. Although the average listing occupancy rate was lower due to the increased market thickness, occupancy rate for those popular listings did not experience statistically significant change (see Online Appendix G for details). Without data on matching quality, one cannot fully examine how consumer surplus would change as a result of thickness change, which we leave for future research.

8. Conclusion

Matching market equilibrium often depends critically on the shape of the matching function—whether it exhibits increasing, constant, or decreasing returns to scale—which is perhaps best tested empirically. However, answering this question is challenging, because market size is typically determined endogenously, making causality difficult to establish. We address this challenge by exploiting an exogenous shock to market thickness caused by a one-time migration of listings from other platforms, which gave rise to a quasiexperimental design.

We find several interesting results. Thicker markets are not necessarily associated with higher matching rates. On the contrary, we find that an increase in market thickness may lead to a lower matching probability for both travelers and hosts, keeping search technology and other factors unchanged. In our context, the platform lost 5.6% of potential matches each day because of the increased market size. This effect can be attributed to increased search friction. We find that facing a larger number of options, travelers search more. Because the matching process is not instantaneous, both travelers and listing owners end up conversing with more potential partners before they can reach a decision, at which point the other party may not be ready to commit yet or may be no longer available. As a result, travelers are forced to quit their search process and seek outside options. The

effect of increases in market size on matching rate is even more prominent when the matching deadline is close.

Our findings have important implications for both theory and practice. They highlight the need for carefully designed empirical research to measure the causal impact of market thickness on matching rates. They also call for careful evaluations of the properties of matching functions and their implications in modeling work. Practically, peer-to-peer marketplace operators should pay close attention to search frictions as markets grow and consider investments or interventions to alleviate congestions or externalities caused by search frictions. This is particularly important when the market exhibits high degrees of heterogeneity and when needs for matching are imminent.

Of course, our study is not free of limitations, which provide several opportunities for future research. First, we do not have access to granular search data. Such data would allow a more in-depth analysis of travelers' search processes, which might offer insights on how platform operators can help users avoid extensive and inefficient searches and find a match quickly. Second, we analyze matching rate, but we do not have direct measures of matching quality (e.g., satisfaction after matching). It would be interesting to investigate whether the quality of matches is better in larger markets, despite a decrease in matching probability. Third, we observe only transacted prices but not listing prices, and therefore, we are not able to fully examine effects of market thickness on prices or the potential effects of supply-side price changes on matching rates and search intensity. Instead, we measure the overall effects of the market thickness on matching rates, search intensity, and (transacted) prices. Indeed, it would be interesting for future research to measure the effect on listing price and decompose the overall effects of the market thickness to direct and indirect effects by estimating the interplay of prices, search, and matching rates. To disentangle the direct and indirect effects, one would potentially need multiple instruments, because prices, search, and matching rates are determined through a system of simultaneous equations. Last, this paper measures the short-term effect after the increased market thickness. In the long run, hosts may decide to charge lower prices after realizing that the conversion rate is low, become less active, or even exit the platform. The platform operator may decide to invest in advanced search technology or provide incentives to keep participants active. There will be a different equilibrium.

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Endnotes

¹ As in most travel reservations, there is a chance that the parties involved may cancel the reservation. Unfortunately, we do not have access to the cancellation data. The focus of the analysis is on the initial matching rate, and most often, cancellations are caused by unforeseen reasons after the initial matching has taken place.

² The results are consistent if we use other cutoffs, such as the 75th or 95th percentile. To avoid treating markets with trivial numbers of new listings in the four-day period as experiencing a discontinuous increase in thickness, we require that the market has to have a minimum of three new listings amounting to at least a 1% increase relative to the number of existing live listings at the time in the market. The results are consistent, however, if we do not impose this additional requirement.

³ Alternatively, one could measure the increase in thickness in terms of listing-days. It has a very high correlation (0.9952) with that calculated using number of listings. As a result, using the number of listing-days leads to almost identical results.

⁴ To ensure clean definitions of “before” and after periods, searches that started during the before period but ended during the after period are excluded from the sample.

⁵ For example, with more options available in the primary market of interest, a traveler may be less likely to search other markets. We indeed find evidence consistent with this conjecture.

⁶ The estimation results are robust to alternative cutoffs: 10 and 25.

⁷ The results are robust to alternative thresholds at the 50th and 90th percentiles.

⁸ The results are consistent if we instead focus on 30 or 90 days before and after the migration of listings.

⁹ We verify that the results are not affected by the nonpresence of an Easter trip during the before period. Easter contains only 3 days. If we exclude trips that contain the Easter holiday, the results are consistent.

¹⁰ The number of prior trips is measured as starting from January 2014, because we only have access to traveler inquiry data in 2014.

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